

Temporal Video Burst-Based Smart Attendance System Using Deep Face Recognition and Interval Voting

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1 Introduction

Traditional attendance systems based on single images are unreliable due to occlusion, pose variation, and transient visibility. To address these issues, a temporal video burst-based approach is proposed, where short video clips are captured at regular intervals and analyzed using deep face recognition techniques.

2 System Overview

The system consists of two main phases:

- Enrollment Phase (Offline)
- Attendance Phase (Online)

3 Enrollment Phase (Offline)

This phase is performed once before deployment.

3.1 Step 1: Data Collection

- Capture 10–15 images per student using a smartphone or webcam.
- Ensure variation in pose, lighting, and expression.
- Face should occupy approximately 40

3.2 Step 2: Face Detection

- Use a pretrained face detection model such as MTCNN or RetinaFace.
- Detect bounding boxes for faces.

3.3 Step 3: Face Cropping and Alignment

- Crop the face region using bounding box.
- Align face using landmarks (eyes, nose).
- Resize to 160×160 pixels.

3.4 Step 4: Embedding Extraction

- Use pretrained FaceNet model.
- Extract embedding vector (128-D or 512-D).

3.5 Step 5: Database Creation

- Store embeddings with student ID.
- Optionally compute mean embedding per student.

4 Attendance Phase (Online)

4.1 Step 1: Video Burst Capture

- Capture 3-second video bursts.
- Repeat at fixed intervals (e.g., 5 times per lecture).

4.2 Step 2: Frame Extraction

- Convert video to frames using OpenCV.
- Sample frames (e.g., 10 FPS).

4.3 Step 3: Face Detection

- Detect multiple faces in each frame.

4.4 Step 4: Face Cropping and Alignment

- Crop detected faces.
- Align and resize.

4.5 Step 5: Embedding Extraction

- Extract embeddings using FaceNet.

4.6 Step 6: Embedding Matching

- Compare with enrolled embeddings using cosine similarity.
- Apply threshold for identification.

4.7 Step 7: Interval Presence Estimation

- Count detections per student in an interval.
- Mark present if count exceeds threshold.

4.8 Step 8: Interval Voting

- Repeat for all intervals.
- If present in ≥ 3 intervals, mark as present.

4.9 Step 9: Attendance Storage

- Store results in CSV file.

5 Algorithm

For each student:

Capture images

Detect and crop faces

Extract embeddings

Store in database

For each interval:

Capture video burst

Extract frames

For each frame:

Detect faces

Crop and align

Extract embedding

Match with database

Update presence count

For each student:

If presence ≥ 3 intervals:

Mark Present

Else:

Mark Absent

Save results to CSV

6 Tools and Libraries

- OpenCV – video processing
- MTCNN / RetinaFace – face detection
- FaceNet – face recognition
- NumPy – numerical operations
- Pandas – CSV handling

7 Code Resources

- FaceNet: <https://github.com/davidsandberg/facenet>
- MTCNN: <https://github.com/ipazc/mtcnn>
- RetinaFace: <https://github.com/serengil/retinaface>
- OpenCV: <https://opencv.org>

8 Conclusion

The proposed system improves attendance reliability by using temporal aggregation and interval voting. It effectively handles occlusion, pose variation, and transient visibility in real classroom environments.